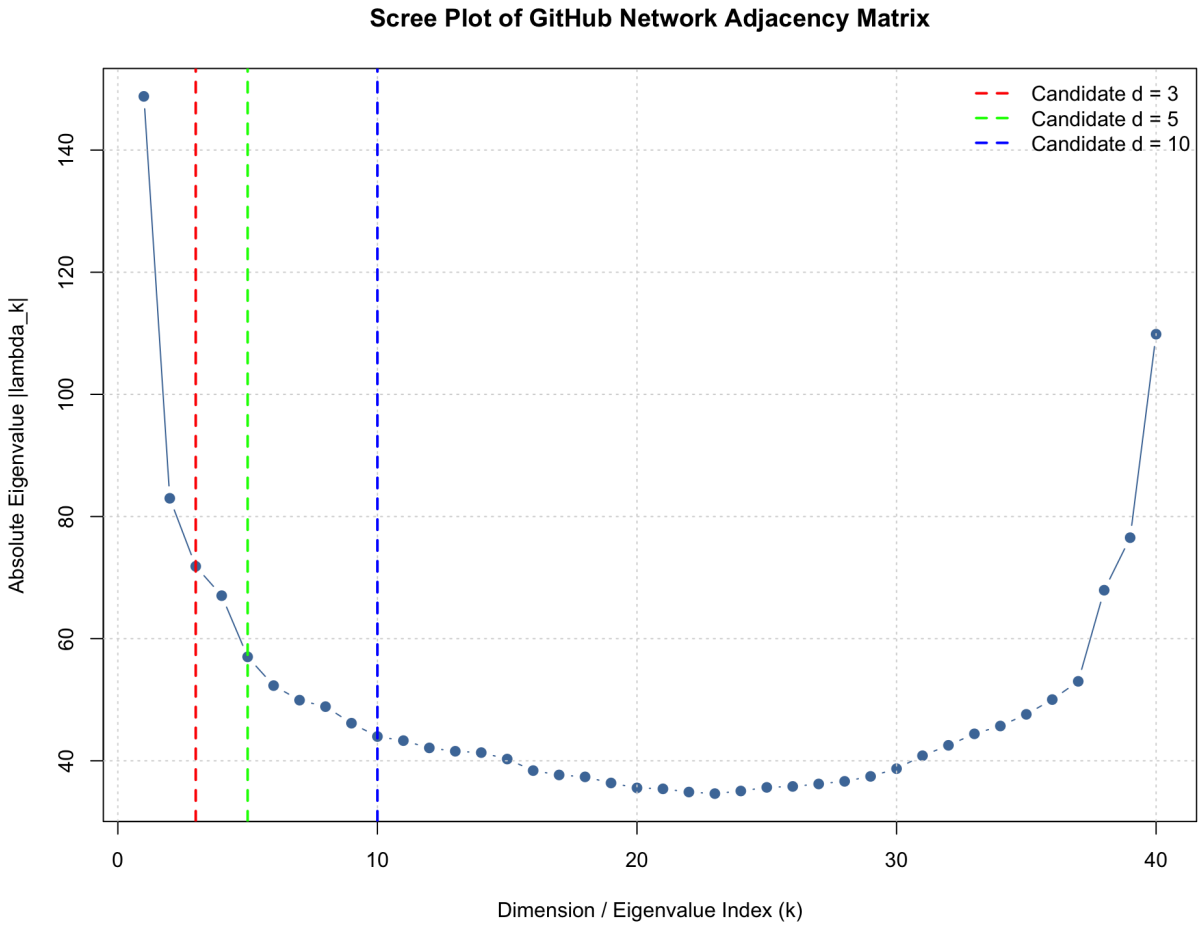
Question 04

A goal of ours is to determine whether we can predict whether someone is a web or machine learning developer. To do so, we constructed a dataset where each node is an observation and each predictor is derived from network characteristics such as degree, clustering coefficient, betweenness centrality, eigenvector centrality, and spectral embedding coordinates derived from singular value decomposition. We then fit a classification model (regularized logistic regression (Lasso)) to predict whether the node is a web or ML developer.

Scree Plot

To decide on the right number of dimensions or the correct number of coordinate columns we want our singular value decomposition (and ultimately our adjacency spectral embedding) to output, we look at a scree plot which ranks every possible dimension by how much its structural weight or importance carries. The scree plot indicates that d=5 (the green line) has a strong elbow and captures the majority of the network's structural variance without making the model too complex.



Results from Lasso

Since we are predicting ML developer (ML = 1, Web = 0), a positive coefficient means that the feature strongly predicts a ML developer while a negative coefficient indicates that the feature strongly predicts a web developer.

The strongest predictor of ML developers is pagerank (7.9772) indicating that despite ML developers having fewer connections than web developers, the connections they do have are with highly influential and important individuals within the network.

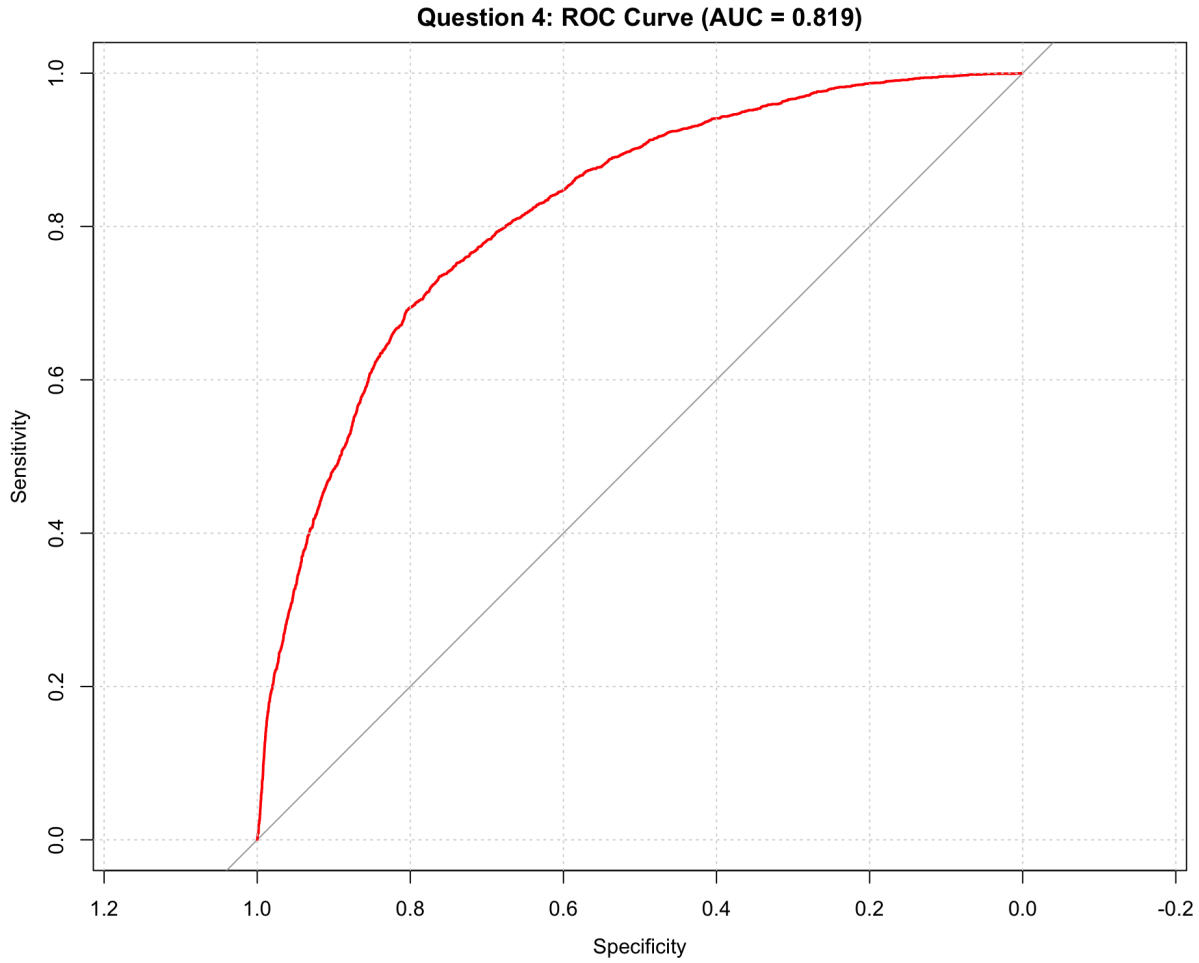
In contrast, the strongest predictor of web developers is degree (-7.3664) indicating that web developers tend to have many connections across the network. Moreover, the high eigenvector coefficient (-4.2007) strongly predicts web developers suggesting that web developers tend to be connected to tightly knit groups. Similarly the negative betweenness coefficient (-1.8915) indicates that web developers tend to act as bridges between different sub-communities.

The five dimensional coordinates from the adjacency spectral embedding shows that the lasso model dropped dimension 1 which is strongly correlated with our degree metrics (degree and pagerank). The higher latent dimensions remain in the model indicating that the latent spatial coordinates derived from the ASE are important to predicting ML and web developers. In fact, the strongest predictor among the latent predictors, dimension 5 (-3.1728), strongly pushes toward web developers while the next strongest latent predictor, dimension 4 (1.9375), pushes toward ML developers.

Predictor	Coefficient
(Intercept)	-2.2814
degree	-7.3664
pagerank	7.9772
clustering	0.0337
betweenness	-1.8915
eigenvector	-4.2007
ASE Dimension 1	.
ASE Dimension 2	-1.0336
ASE Dimension 3	1.1249
ASE Dimension 4	1.9375
ASE Dimension 5	-3.1728

ROC Curve

The ROC curve shows that the network structure is a highly reliable predictor of developer type (with an AUC = 0.819).



Precision-Recal AUC

The ROC curve shows that the network structure is a highly reliable predictor of developer type, achieving an AUC of 0.819. Furthermore, because web developers outnumber machine learning developers roughly 3-to-1, we also evaluated the Precision-Recall AUC (PR-AUC). The model achieved a PR-AUC of 0.61 (substantially higher than the ~0.25 baseline of a random guess), confirming that the network features successfully and consistently identify the minority machine learning class despite the skewed dataset.

